

A NEW SUPPORT STRATEGY FOR REDUCING WASTE IN ADDITIVE MANUFACTURING

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ABSTRACT

In additive manufacturing, fused filament fabrication (FFF) printers are widely used because of their low cost and simplicity. However, such printers still suffer from redundant support material waste when printing objects with overhangs, holes or edge features. In this paper, a new support strategy considering printable threshold overhang angle (PTOA) and the longest printable bridge length (LPBL) is proposed. Based on this, the to-be-supported regions will have the least contact area and support waste. The characteristics of LPBL and PTOA are fully considered for saving more material. A T-shape part is used to verify the proposed strategy and the results show that this support strategy can considerably reduce material waste compared with conventional strategies, enabling AM to be a more environmentally friendly and sustainable manufacturing technique.

Keywords: Additive manufacturing, Longest printable bridge length, Printable threshold overhang angle, Support.

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1 INTRODUCTION

Additive manufacturing (AM) has been developed rapidly in the past 30 years. The most widely used AM technique is fused filament fabrication (FFF). The manufacturing process of FFF starts from the bottom of a part and carries on in a layer-by-layer manner to the top. This leads to problems for overhangs that cannot be printed since there is no supporting layer beneath them [1,2]. Therefore, sacrificial external support structures are needed for assisting overhanging features in 3D printing processes. After fabrication completes, supports are chemically or mechanically removed and discarded, thus wasting the material used. This usually involves human intervention and therefore is time-consuming and expensive.

Many studies have been carried out for reducing support material waste [3]. Seeking better structures such as cellular and lattice structures is a common way for reducing the support usage and cost. As the advantage of their low volume fraction, cellular structures provide opportunities to reduce the volume of support materials and allow for easy removal of support structure as well as build time. Vaidya and Anand [4] employed Dijkstra's shortest path algorithm [5] to generate cellular support structures. Strano et al. [6] proposed a new approach which applies a new optimisation algorithm to use pure mathematical 3D implicit functions for the design and generation of the cellular support structures including graded supports. In their method, different cellular structures can be easily defined and optimised, especially graded structures providing less support where requiring less weight support and more robust support elsewhere. Tree-like support structure methods were presented by [7] and [8].

Another method to save cost in terms of support is by printing supports with different materials which are soluble or sacrificial in some way. This means when printing a part, using the part material to fabricate the final part while using another sacrificial material (either easier to remove or cheaper than the part material) to print the support. In FFF processes, soluble support materials have been used extensively, with preference given to polymer feedstocks that are soluble in relatively benign solvents such as water (poly(vinyl alcohol)) and limonene (high impact poly(styrene)). Another common combination is a part printed with acrylonitrile butadiene styrene (ABS) while a sacrificial support structure printed with soluble polylactic acid (PLA), which is later selectively removed by immersing the 3D printed structure in a solution of isopropyl alcohol and potassium hydroxide to remove the soluble support structures (PLA), while the ABS component will keep intact [9]. Hopkins et al. [10] proposed a support material containing the first copolymer and the polymeric impact modifier. The new support material can effectively resist cracking or breaking, and can be removed easily and quickly. In an alkaline aqueous solution, the material can be dissolved rapidly. In the study of Pekkanen [11], charged poly(ether ester)s with tailored rheological and mechanical properties were conducted as novel materials for extrusion-based AM at low temperatures. Ni et al. [12] also conducted some research on Poly(vinyl alcohol) (PVA) which can act as water-soluble supports in 3D printing processes. The method of adopting two materials in one fabrication process can also be used in metal 3D printing technologies [13,14].

In this paper, a new support strategy is proposed for reducing support waste, considering printable threshold overhang angle (PTOA) and the longest printable bridge length (LPBL). Taking the advantage of PTOA and LPBL, the to-be-supported regions will have the least contact area and support waste. A T-shape part is used to verify the proposed strategy.

2 PRINTABLE THRESHOLD OVERHANG ANGLE AND LONGEST PRINTABLE BRIDGE LENGTH

For easier understanding the proposed support strategy, a background about printable threshold overhang angle (PTOA) and longest printable bridge length (LPBL) are introduced first.

2.1 Printable threshold overhang angle (PTOA)

In additive manufacturing processes, each layer needs to have another layer beneath it for supporting printability due to the layer-by-layer nature. However, overhangs sometimes can also be manufactured without support layers when the overhang angle is larger than some threshold [15], due to material properties. PTOA means an overhang with the lowest angle that can be fabricated without adding support during the deposition process. In this paper, the overhang angle (α) is defined as the angle between the x-y plane and the overhang surface tangent in the x-z (or y-z) plane (see Fig. 1). Generally, PTOA is set at 45° in most printers. Some also take a test first to determine the angle size as it may be different according to printers and materials [16-18]. Also, print process parameters are influential to PTOA according to our previous research [15]. Fig. 2 shows some printed overhangs in different angle sizes with the smallest printable size (the cross-section in the x-y plane is a square of 0.8×0.8 mm). As the aim of PTOA test in this study is for support, the surface quality of the printed overhang pillars is not the focus, once the overhang pillar in the smallest printable size is able to be self-supported, the corresponding overhang angle size is the PTOA. In this paper, the characteristics of PTOA is fully considered for minimizing support usage.

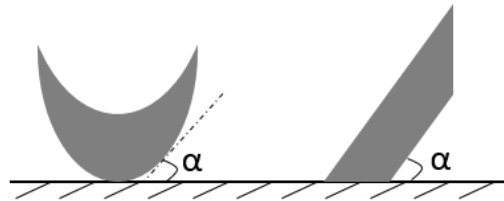


Figure 1: Definition of overhang angle (β) in this paper

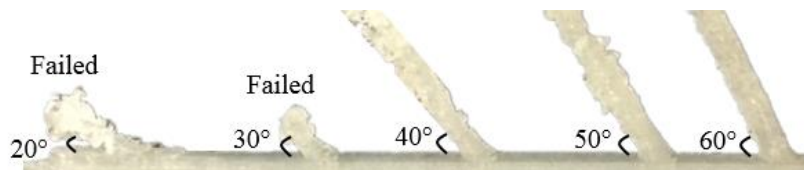


Figure 2: Printed overhangs in different overhang angle sizes

2.2 Longest printable bridge length (LPBL)

LPBL means that a bridge with a longest length a 3D printer can build (with satisfied finish quality) without support structure underneath it. During a fabrication process, there are a lot of factors (e.g. print temperature, print speed, solidification speed) that may influence LPBL. Fig. 3 shows some printed bridges with different lengths under different process parameters and the same print path direction. It clearly shows that LPBL can be achieved in various lengths when changing process parameters. At the same time, the characteristics (e.g. strength, solidity, quality) of printed part can also be influenced by these process parameters [19,20]. For satisfying the requirements of a product, the process parameters will need to be certain in a range. Our previous work shows more details of LPBL [21]. The satisfied LPBL can be experimentally tested according to the required process parameters. In this paper, LPBL will be integrated into our proposed support strategy for reducing material consumption.

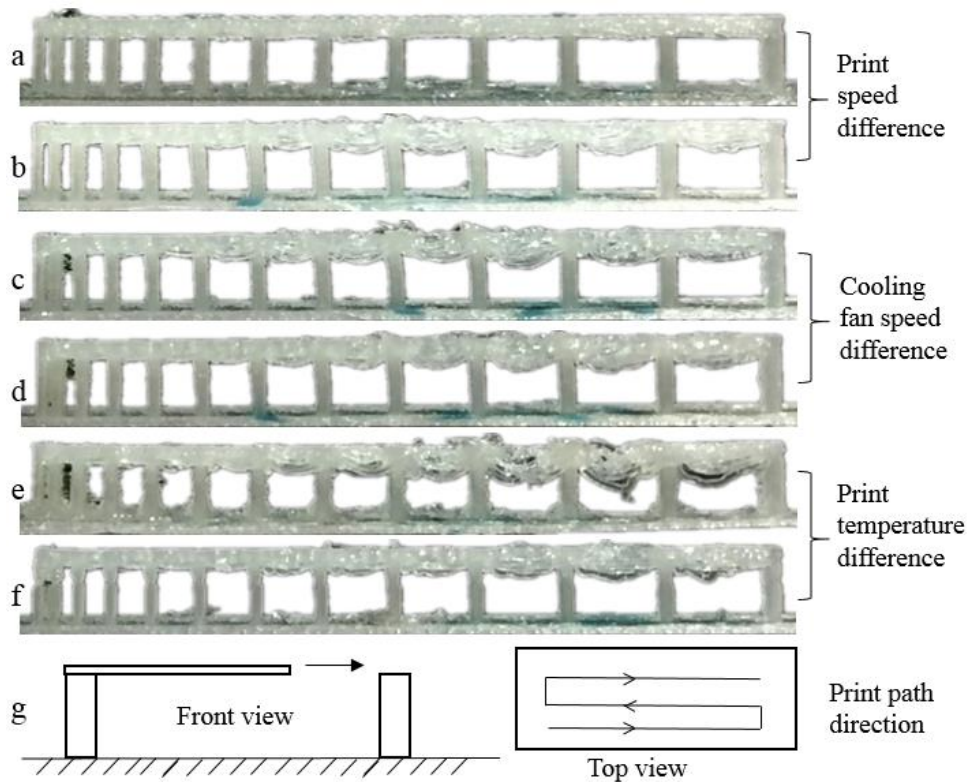


Figure 3: Samples printed in different process parameters

3 GENETATING SUPPORTS BASED ON LPBL AND PTOA

In the previous section, LPBL and PTOA have been introduced. Based on this background, a support generation strategy by fully taking the advantage of LPBL and PTOA is proposed as follows. The following illustration takes the T-shape model shown in Fig. 4 as an example. The support structures follow three rules.

- The distance between support points (see Fig. 5(a)) should be the LPBL, which can save as much material as possible.
- The PTOA should be used as the threshold angle for generating supports. In Fig. 5, 40° was used as PTOA size. 40° is just an example, the exact value of angle size should be determined by tests.
- The shortest support length should be chosen with the aim of using least material. For example, when generating support in the situation of strut a or b in Fig. 5(a), strut a should be used rather than b as a is shorter than b.

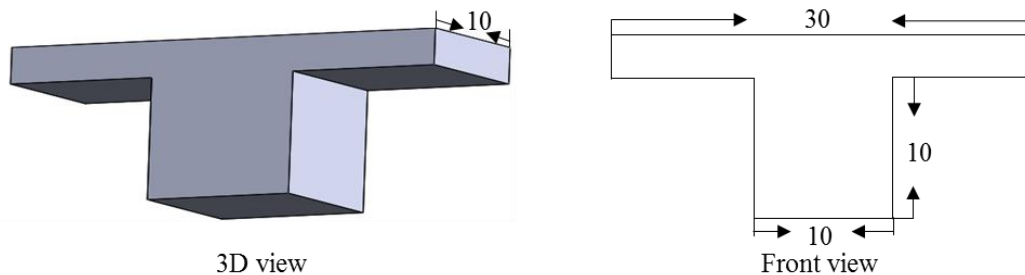


Figure 4: T-shape model and its dimensions

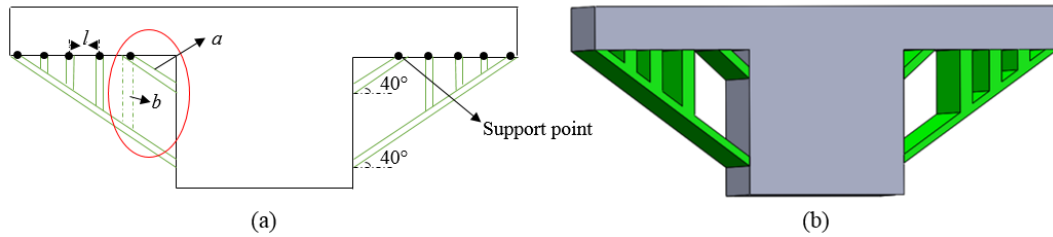


Figure 5: (a) 2D illustration of support generation; (b) Final generated supports (green regions)

4 DEMENSTRATION AND DISCUSSION

To verify the proposed support generation strategy, this T-shape model was printed on a Kossel Delta 3D printer, using optimized print strategy. The build area shape of the 3D printer is circular with maximum width of 180 mm, maximum depth of 180 mm and maximum height of 300 mm. The nozzle diameter of this printer is 0.4 mm. Polylactic Acid (PLA) was used for printing parts.

Before generating support structures, LPBL and PTOA are tested first with the process parameters of print temperature of 190°C , print speed of 20 mm/s, cooling fan speed of 250 RPM, fill density of 30% and layer thickness of 0.2 mm. According to the experimental results, LPBL can be achieved at 2.0 mm when setting the deformation tolerance within 0.15 mm and PTOA can be achieved at 40° under these print parameters in this Kossel Delta printer. Therefore, the support structures were generated based on the corresponding LPBL and PTOA.

For comparison, another two support methods provided by slicer software Cura 15.04 were also used for fabricating the same part. The results of these three support methods are shown in Fig. 6. As can be seen, our new support generation method only consumes 0.3g support material for the “T” part, while line and grid support methods spend 2.1g and 2.3g respectively. In addition, it is easier to remove the support from our method than line and grid methods. In terms of the surface quality after removing the supports, all these three methods leave minor marks on the supported surface. However, our method leaves the least mark area on the surface as our method is based on LPBL and has the smallest contact area. This may help reduce the effort on post-processing and it is easier to be dealt with. However, the final printed qualities (strength, modulus, etc.) may be affected by using our strategy, more investigations will be carried out for further improvement in the future.

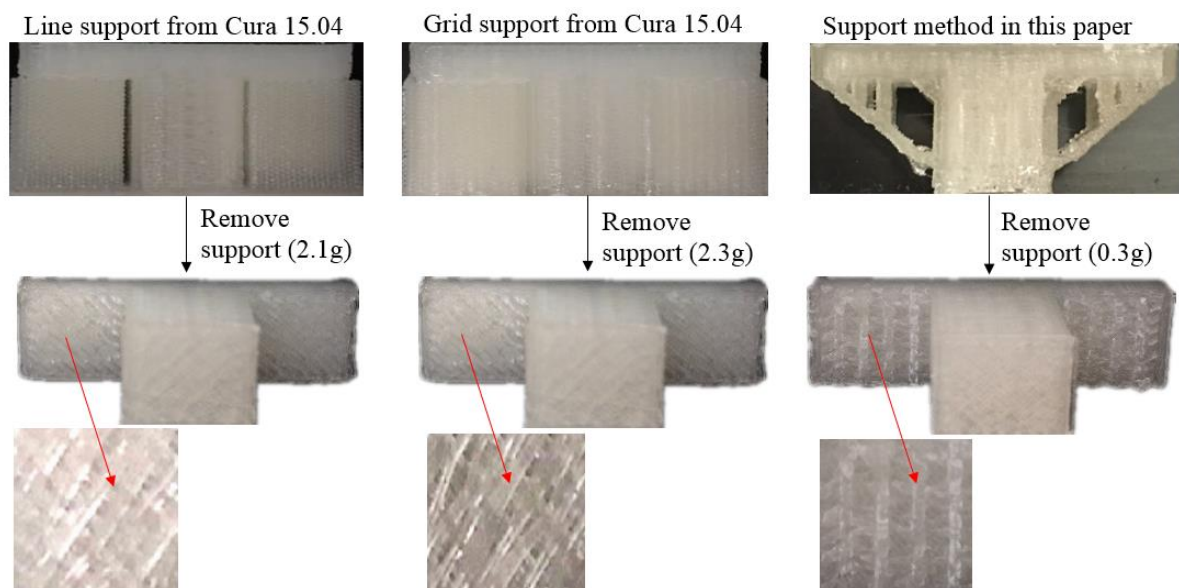


Figure 6: Comparison between different support methods

5 CONCLUSIONS

In this paper, a new support strategy for AM considering LPBL and PTOA is proposed. The characteristics of LPBL and PTOA are fully considered for reducing the support usage. T-shape parts were fabricated to verify the proposed strategy by comparing them with the general methods (Grid and Line methods from Cura 15.04). The results show that the proposed strategy can considerably reduce support material consumption, enabling AM to be a more environmental friendly and sustainable manufacturing technique. In addition, the proposed support strategy has better finished surface quality than general support methods.

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